

Fire-Works: An Interactive System Based on Breathing Patterns to Support Social Interactions Among the Oldest Old Living in Long-Term Care Facilities

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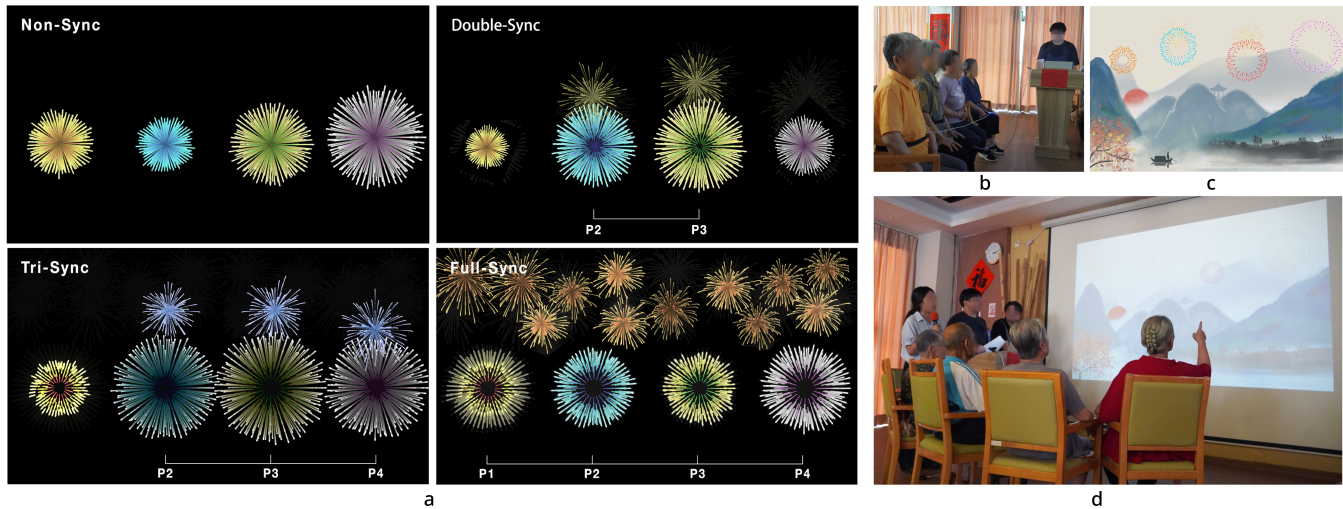


Figure 1: Fire-Works system introduction and user experience: (a) Visualization of the system phases. Non-Sync scenario: participants visualize their breathing patterns on the screen. Double-Sync scenario: two participants among four manage to synchronize their breathing. Tri-Sync: three participants among four manage to synchronize their breathing. Full-Sync: all four users achieve synchronization. (b) Participants wore a breathing belt around their abdomen; research assistants stood nearby. (c) A Chinese landscape painting as a background image during workshops. (d) Four participants were seated in front of a large screen, interacting with the visual feedback.

Abstract

For the "oldest old" living in Long-Term Care (LTC) facilities, declining physical and cognitive abilities often make it challenging to engage in social interactions with other residents. This paper presents a bio-feedback system as a technology probe that allows four users

to visualize their breathing and synchronize with each other to trigger social interactions through their bodies. We conducted an exploratory study in an LTC facility through five interactive workshops that included twenty older adults, 16 of them aged 80 and above. Our research reveals that this interactive system employs easy breathing techniques to empower elderly residents with a sense of control, allowing them to engage in verbal and non-verbal social interactions. This interactive breathing-based system shows great potential as a technology-mediated intervention deployable in LTC environments.

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CCS Concepts

• **Human-centered computing** → *Human computer interaction (HCI); Accessibility systems and tools.*

Keywords

Long-Term Care facilities, older adults, breathing patterns, interactive system, social interactions

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1 Introduction

Most older adults living in LTC facilities, particularly those "oldest old" (people over 85)[31], decline in physical and cognitive abilities, often making it difficult to engage in social interactions with other residents effectively [3]. Although regular recreational activities like singing, dancing, and crafting can offer opportunities [12], some older adults may even limit their social interactions due to fear of becoming a burden or lacking confidence in their communication skills [23].

Previous studies have introduced technology-mediated interventions such as the systems that bring scenery pictures and interesting photos into LTC facilities [10, 11, 19, 20] or interactive tangible devices to encourage older adults to tell their stories [18, 21]. These systems generally aim to provide "tickets to talk" [2] to enhance interactions through verbal communication. However, these studies overlook the reality of declined verbal communication abilities among older adults, particularly the "oldest old." Therefore, it is critical to develop more designs concerning elderly residents with difficulties in verbal expression and physical movements. Some studies, such as Vita designed for the dementia [14], which creatively incorporated everyday sounds into a pillow, and Tovertafel, a series of interactive games projected onto a table [1], adopted simple interactive motions like listening and moving hands, which shows the great potential of non-verbal behaviors in creating more opportunities for social interactions among elderly residents.

Inspired by bio-feedback systems, this research explores how unique non-verbal behaviors — inner physiological movements, can be utilized to facilitate social interactions. Recent research shows that interactive breath-based systems enhance interpersonal connections through bio-signal sharing and breathing synchronization [26, 28]. For example, JeL is a VR system that achieves interpersonal synchronization through breathing by mapping breath movements to the motion of virtual jellyfish [7]. When users synchronize their breathing, a glass-sponge-inspired sculpture begins to grow. By encouraging users to synchronize their breathing, it promotes interaction and alignment with others' bodies. This mutual adaptation creates a positive feedback loop, guiding individuals to focus on their internal bodily states and their relationships with others, thereby supporting empathy and a sense of connectedness [25, 26].

Furthermore, breathing, as a controllable bio-signal, allows for semi-unintentional expression, reacting automatically to a person's state but can be regulated with conscious awareness [9]. This trait gives to breathing more significant interactive potential than other physiological signals, opening new avenues for designing systems

for the "oldest old." Previous studies have leveraged this characteristic to create innovative designs. For example, Bubble Beats [15] is a game designed for children that integrates breathing exercises with musical rhythms. Similarly, ChillFish [30] is a calming biofeedback game developed for children with ADHD, aiming to improve attention retention by combining breathing exercises with a video game. These designs primarily focus on engaging users, particularly children, in breath training, thereby promoting their overall well-being.

In this research, we are particularly interested in exploring the social potential of breathing signals across a broader population. We developed *Fire-Works*, a breath-based interactive system allowing four users in one group to visualize their breathing patterns as fireworks and experience different scenarios based on their level of breathing synchronization (Figure 1). We conducted five interactive workshops in an LTC facility to understand older adults' perceptions and acceptance of it and the potential issues that may arise during the implementation. The workshops were conducted with 20 elderly residents divided into five groups. 16 participants were the "oldest old" above 80 years old. The workshops were followed by interviews with all participants and a discussion with three social workers and two volunteers.

Our work provides three primary contributions: 1) We developed *Fire-Works*, a breath-based interactive system targeting the "oldest old." 2) We deployed *Fire-Works* within an LTC facility, providing them with a new tool for enhancing social interactions among elderly residents. 3) We contribute insights into how such a group perceives and interacts with *Fire-Works*.

2 Fire-Works: A breath-based interactive system

2.1 System Design

We designed a multi-user interaction system based on breathing pattern detection and visualization to offer a new opportunity for elderly residents to engage in social interactions with others. We developed *Fire-Works*, a system allowing four users to experience it simultaneously, and divided into two different phases.

In the first phase, users can visualize their breathing patterns and those of others. We chose fireworks as a visual metaphor for breathing visualization to express abstract breathing data and help elderly users to better understand the system. Fireworks commonly evoke celebration, festivity, and joy, conveying that "each breath is a celebration of life." The expansion or contraction of the firework corresponds to the user's inhalation or exhalation. We define this scenario as Non-Sync.

In the second phase, users are encouraged to synchronize their breathing patterns with others. Different firework effects are triggered when two, three, or four users synchronize their breathing, defined respectively as Double-Sync, Tri-Sync, and Full-Sync. Due to the semi-unintentional feature of breathing, breath synchrony occurs frequently in group activities, promoting the shared experience of an intimate connection between people [6, 22, 29]. Encouraging users to synchronize their breathing promotes interaction and alignment with others' bodies [26]. In addition, to increase the visual effect and user engagement, we use a Chinese landscape painting as a background image.

2.2 System Implementation

Our system consists of two main components: four breathing detection belts (Elastech) to detect breathing patterns through motion sensors and a program developed by Processing⁴ to receive and display breathing data as fireworks (Figure 2). The program maps breathing data to changes in the radius of the fireworks.

A calibration occurs during the first 15 seconds, dealing with individual differences in breathing intensity. This phase captures each user's breath amplitude, using the peaks and valleys to map the data to a predefined radius range. After calibration, the system compares each user's breathing frequency, amplitude, and phase in real-time to determine if multiple users' breathing is synchronized. When synchronization is achieved, the system triggers Double-Sync, Tri-Sync, and Full-Sync scenarios corresponding to the number of synchronized users. This design encourages elderly participants to interact with each other through breathing synchronization.

3 Methods

We conducted an exploratory study, which included five interactive workshops with 20 participants in an LTC facility and a discussion with stakeholders, to understand i) the acceptance and interest in *Fire-Works* of older adults, ii) how social interactions take place while experiencing the system in a real context, and iii) the role of *Fire-Works* in promoting social interactions. We obtained approval from the Ethics Review Board of the institution to which the authors belong.

3.1 Setup

The study took place in the activity room of an LTC facility in China, which had a projector mounted on the ceiling and a large screen. Initially, four chairs were arranged in a semi-circular formation in front of the screen to ensure participants could see it while observing their peers' reactions. The room's lighting was dimmed to enhance the visibility of the projected images. We also positioned a camera facing the participants in one corner of the room to record the experience and collect video data (Figure 3).

During the workshops, participants took turns using the system. Elderly residents were encouraged to engage with *Fire-Works* and further share their thoughts in a relaxed group setting. While one researcher was a facilitator, another managed the system's operation. Additionally, two researchers quietly stood aside, observing and recording the entire process without interference. They also collected photo, audio, and video data with the participants' permission. Three social workers and two volunteers also helped ensure a smooth execution of the activity.

3.2 Participants

Participant inclusion criteria were as follows: 1) residents assessed by the center as self-sufficient or semi-self-sufficient; 2) no severe cognitive or visual impairments; and 3) having enough energy and willingness to participate in the workshop. The social workers promoted the event within the center and successfully recruited 20 elderly participants who met the criteria to participate in the activity. The average age of the elderly participants is 86, with 16

being 80 years old and above. On the event day, these 20 participants spontaneously formed groups of four. All participants were compensated with a gift for a 25-minute activity time.

3.3 Procedures

Each workshop session followed the following steps: 1) The four participants took their seats and wore the breathing belts. 2) The program began with calibration, where the facilitator guided the users through slow, deep breathing for 15 seconds. 3) The facilitator helped them identify their corresponding fireworks. 4) The participants were then given approximately 120 seconds to explore the system freely. 5) The facilitator guided all participants to synchronize their breathing. 6) The experience concluded. 7) Researchers led the participants to the chat area for an open-ended interview. All interviews lasted approximately 15 minutes each and were carried out face-to-face. With the participants' consent, all interviews were audio-recorded and later transcribed for analysis. Participants were free to interact and move as they wished throughout each step. After the workshop, participants received a printed photo displaying a snapshot of their breathing pattern during the experience.

After completing all workshops, we also discussed with three social workers and two volunteers who participated in the activities. All five stakeholders were well-acquainted with the situation at the center. The interviews lasted 15 minutes, during which we gathered their feedback and suggestions regarding *Fire-Works*.

3.4 Measurements

Our data included 1) observation notes from research assistants, 2) reflective field notes from facilitators, 3) audio recordings of group interviews with participants and stakeholders, 4) video recordings of the workshops, and 5) screen recordings of *Fire-Works* program.

We transcribed and manually cleaned the interview recordings. We grouped and highlighted key statements from the interviews using an inductive coding approach [4]. Additionally, we integrated observation notes from research assistants and reflective field notes from facilitators. Considering some elderly participants have trouble expressing themselves through verbal communication, we further analyzed video recordings to examine non-verbal behaviors [32] of them during the workshops. We used video annotation tools to mark occurrences of eye contact, laughter/smiling, touch, and nodding, as previous literature indicates that these behaviors signal coordination, affinity, and connection in social interactions [5, 13, 16, 17]. Two researchers independently reviewed and annotated the same videos using the same coding scheme. We also recorded the screen during each session to track *Fire-Works* program's operation. By observing the frequency of fireworks in the recorded footage, we assessed the synchronization of participants' breathing within each group. We combined these data to visualize participants' non-verbal behaviors and synchronization patterns on each group's timeline (Figure 4).

4 Results and Discussion

4.1 Behaviors that Emerged with *Fire-Works*

From the video analysis, findings reveal that the 20 elderly participants displayed a range of non-verbal behaviors during their experience with *Fire-Works*, emphasizing eye contact, laughter/smiling,

⁴<https://processing.org/>

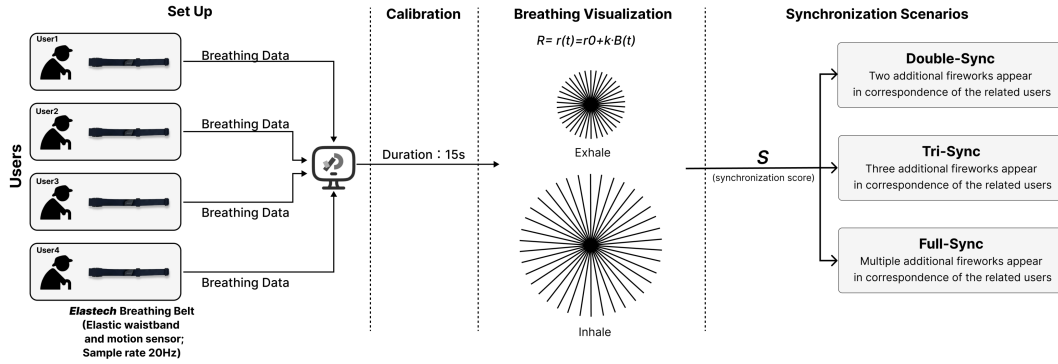


Figure 2: System design, data processing, and visualization scenarios, including four stages: set up, calibration, data visualization and synchronization scenarios

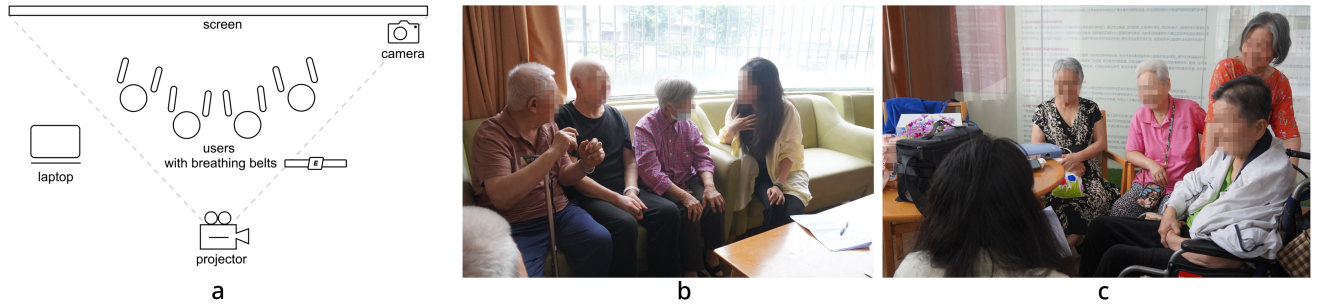


Figure 3: (a) Workshops set up. (b and c) Different groups had follow-up interviews after the workshops.

touch, and nodding. During the calibration phase, all participants focused on the facilitator's guidance and performed slow breathing, completing approximately three breathing cycles within 15 seconds. Once the calibration was completed, four large fireworks appeared on the screen, and some participants visibly smiled at this moment (P1, P7). In the pre-exploration phase (after calibration), the facilitator asked the participants if they had found their fireworks. Initially, most participants concentrated on the screen, trying to match their breathing with the fireworks. After some time, they actively began communicating with the facilitator about which number or color represented their fireworks, with a few additionally pointing at the screen (P13, P14). After each participant had identified their firework, the facilitator encouraged them to explore the system freely. Notably, in a short time, all five groups triggered Double-Sync, and two groups triggered Tri-Sync. After the facilitator explained the cause of the additional small fireworks effect, more participants smiled the next time they triggered it. Participants displayed eye contact, occasional leaning toward each other, and touching. These behaviors varied depending on individual differences and the relationships among them within each group. In the final phase, the facilitator guided the participants in synchronizing their breathing. All participants in each of the five groups followed the facilitator's lead and successfully achieved Full-Sync. Most participants exhibited evident smiles when the colorful fireworks burst across the screen.

4.2 Acceptance and Interest Toward *Fire-Works*

Older adults' decline in physical and cognitive capabilities often leads to stereotypes of them being fragile or dependent and hesitant in engaging with technology [27]. However, through organizing interactive workshops to test *Fire-Works*, we found that older adults were generally quite receptive to it.

For instance, the participants could consciously adjust their breathing to engage with the system. Researchers observed that some participants intentionally took slow, deep breaths to observe changes in the fireworks on the screen, which helped them identify their corresponding fireworks. We also report on the specific behaviors and subjective experiences of the participants. For example, after the 15-second calibration, four large fireworks appeared on the screen, and one participant's expression shifted from confusion to laughter (P6). When asked about her experience during the follow-up interview, she said it felt like she had "lit the fireworks with her breathing." Another participant (P15), upon identifying her firework, excitedly demonstrated to the facilitator how she controlled the firework's size through her breathing. In the post-workshop interviews, when asked, "What do you think of this system?" three participants responded with comments such as "It's high-tech," "Very creative and novel," and "Quite fun" (P1, P5, P6).

These results support the idea that older adults can autonomously control and interact with their breathing. We further reflect on the design of *Fire-Works* to understand how it functions. The system's

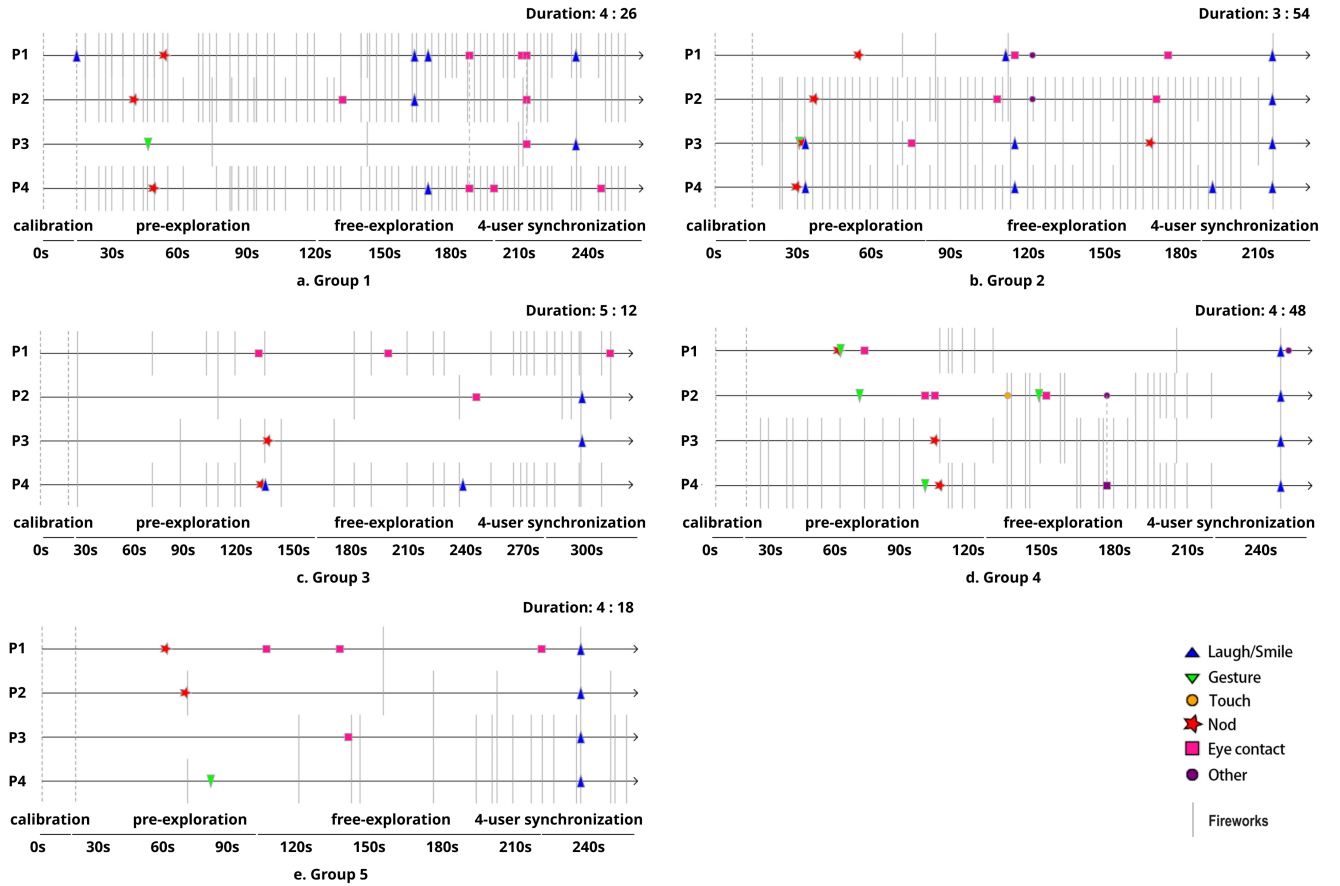


Figure 4: Timeline analysis of non-verbal behaviors and synchronized fireworks of all five groups.

simplicity and intuitive design, relying on a universal and familiar action like breathing, made it more accessible and reduced technological fears. Additionally, a visual metaphor recalling relatable traditional elements effectively transposed an invisible breathing signal into an engaging visual interactive experience.[27], making them receptive on this occasion to engage with technology-mediated interventions like *Fire-Works*.

4.3 The Role of *Fire-Works* in Facilitating Social Interactions

Fire-Works ties breathing patterns to visual fireworks feedback, providing psychological empowerment by allowing older adults to regain control through simple breath-based interactions and fostering unique, non-verbal interactions through synchronized breathing with other residents. Specifically, we further discussed how this system functions in the following aspects.

Shared breathing, shared fate. Breath, as a significant physiological signal, represents an intangible aspect of our inner selves, symbolizing both "life" and each individual's unique existence [8, 24]. Our system visualizes users' breath as "fireworks," allowing them to observe their and others' inner states. In subsequent interviews, participants were asked, "Did you notice the fireworks of others?"

Most participants nodded or responded affirmatively, indicating they had seen them. We also asked, "How did you feel when you could see your own and others' breathing?" and "What was your experience when you synchronized your breathing with others?" Some elderly participants replied, "It was good" or "Quite good" (P3, P7). In the second and fourth groups, participants expressed sentiments such as, "It made us resonate" and "Shared breathing, shared fate" (P8, P15). These results suggest that the elderly participants were interested in "seeing" their own and others' breathing states.

Collaboration sparks fireworks. Producing small fireworks requires participants to synchronize their breathing with others. This synchronization necessitates mutual adjustment of breathing rhythms based on the real-time expansion or contraction of the fireworks. Particularly for triggering Tri-Sync and Full-Sync, participants need to coordinate with companions. As reported by the elderly participants themselves, when asked about their feelings towards *Fire-Works*, they described it as "We are a team" (P12), "We collaborated well" (P11), and "It's like a game; everyone worked together to make the fireworks bloom" (P9). Moreover, the visual feedback provided by the fireworks fosters a sense of shared presence, allowing participants to remain silent without feeling isolated,

which is especially beneficial for those with diminished expressive abilities.

An opportunity to connect with others. Our system provides elderly individuals with an opportunity to initiate and respond to social interactions distinct from simply observing synchronized visuals. When older adults wish to communicate with a particular companion, they can actively adjust their breathing rhythm to synchronize with that companion. The fireworks serve as visual feedback, conveying the information of synchronized breathing to participants. Observing the synchronized fireworks makes participants aware that someone's breathing rhythm matches theirs. This dynamic was demonstrated in the interaction between P14 and P15 during the workshop in the fourth group. During the *Fire-Works* experience, we observed that when the facilitator informed participants that synchronized breathing could trigger small fireworks, P15 noticeably shifted attention to P14's fireworks, actively adjusting their breathing to align with P14. Our visualization results also showed that P14 and P15 frequently triggered synchronized small fireworks over time. In subsequent interviews, when asked about their frequent synchronized fireworks, P15 explained that P14 is his piano teacher at the facility, and he noticed that P14 was particularly happy when seeing the small fireworks. Therefore, P15 aimed to synchronize with P14 to generate more fireworks and enhance P14's enjoyment.

4.4 Perspectives from Social Workers and Volunteers

We discussed about *Fire-Works* with two volunteers and three social workers who participated in the workshops. One social worker noted, "Your design concept is great. If you could add more stages, like a game, and give the seniors more time to engage, I think it would better validate your design." Another social worker added, "You could explore how different group sizes work—what happens with 2, 3, or 4 participants, or even with 10. You could try different themes and see how they produce different effects." They also said, "This activity brings people together, allowing them to feel each other's breathing slowly. I believe this could strengthen group cohesion." Although we had not named the event, one of the volunteers coordinating the activities remarked after its conclusion, "Thank you so much for hosting today's 'Breathing in Harmony, Sharing the Rhythm of Life' activity." These results suggest that deploying our system in real-world environments is a foreseeable future.

4.5 Limitations and Future Work

In this research, the female-male ratio in our sample was 15:5. The unbalanced gender distribution and the influence of a specific cultural background may have a particular impact on the results. In the future, we plan to expand *Fire-Works* to more LTC facilities to balance gender distribution and reduce the effects of cultural background differences. Regarding equipment selection, we will explore more non-invasive ways to collect breathing data, such as infrared sensors, computer vision recognition technology, etc., to improve comfort and user experience. In addition, we tested *Fire-Works* in the LTC facility for a short period, and we are aware that it remains unclear whether the system can sustain long-term interest and social benefits. Based on feedback from participants

and stakeholders, the interaction design of *Fire-Works* at this stage can be further enhanced to better engage elderly residents and encourage prolonged use. Specifically, incorporating varying user group sizes and introducing additional visualization and interaction modalities may help increase user participation. Moving forward, we plan to refine these improvements, incorporate baseline measurements, and conduct a long-term evaluation to more accurately assess the system's impact. Moreover, we will consider adopting multi-sensory design methods such as sound or haptic feedback to improve accessibility for older adults with different disabilities.

5 Conclusion

We presented *Fire-Works*, a breath-based interactive system designed as a technology probe. We conducted five interactive sessions with older adults in an LTC facility while gathering stakeholder insights. Our findings indicate that the elderly participants showed high levels of acceptance and interest in *Fire-Works*, which allows them to engage socially beyond their physical limitations, offering them a sense of control and fostering social interactions through non-verbal behaviors. Further research is needed to refine the system's design; nevertheless, taking this work as a foundational first step, we aim to explore how technology-mediated interventions can be introduced into LTC facilities and be extended to a broader range of older adults, especially the "oldest old," to foster social interactions, consequently supporting their well-being.

References

- [1] Hester Anderiesen. 2017. *Playful Design for Activation: Co-designing serious games for people with moderate to severe dementia to reduce apathy*. Dissertation (TU Delft). Delft University of Technology. doi:10.4233/uuid:ebef0fa-46fe-4947-86c1-c765a583770a
- [2] Mark Blythe, Peter Wright, John Bowers, Andy Boucher, Nadine Jarvis, Phil Reynolds, and Bill Gaver. 2010. Age and experience: ludic engagement in a residential care setting. In *Proceedings of the 8th ACM Conference on Designing Interactive Systems* (Aarhus, Denmark) (DIS '10). Association for Computing Machinery, New York, NY, USA, 161–170. doi:10.1145/1858171.1858200
- [3] Robin P Bonifas, Kelsey Simons, Barbara Biel, and Christie Kramer. 2014. Aging and place in long-term care settings: Influences on social relationships. *Journal of Aging and Health* 26, 8 (2014), 1320–1339.
- [4] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [5] Darren David Chadwick and Tracey Platt. 2018. Investigating humor in social interaction in people with intellectual disabilities: A systematic review of the literature. *Frontiers in psychology* 9 (2018), 1745.
- [6] Erwan Codrons, Nicolò F Bernardi, Matteo Vandoni, and Luciano Bernardi. 2014. Spontaneous group synchronization of movements and respiratory rhythms. *PLoS one* 9, 9 (2014), e107538.
- [7] John Desnoyers-Stewart, Ekaterina R. Stepanova, Philippe Pasquier, and Bernhard E. Riecke. 2019. JeL: Connecting Through Breath in Virtual Reality. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–6. doi:10.1145/3290607.3312845
- [8] Norman Farb, Jennifer Daubenmier, Cynthia J Price, Tim Gard, Catherine Kerr, Barnaby D Dunn, Anne Carolyn Klein, Martin P Paulus, and Wolf E Mehling. 2015. Interoception, contemplative practice, and health. *Frontiers in psychology* 6 (2015), 763.
- [9] Yuan-Ling Feng, Mingyue Gao, Chih-Heng Li, Zhihao Yao, and Haipeng Mi. 2024. ExBreath: Explore the Expressive Breath System as Nonverbal Signs towards Semi-unintentional Expression. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 134, 7 pages. doi:10.1145/3613905.3650870
- [10] William Gaver, Andy Boucher, John Bowers, Mark Blythe, Nadine Jarvis, David Cameron, Tobie Kerridge, Alex Wilkie, Robert Phillips, and Peter Wright. 2011. The photostroller: supporting diverse care home residents in engaging with the world. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing

- Machinery, New York, NY, USA, 1757–1766. doi:10.1145/1978942.1979198
- [11] William W Gaver. 2006. The video window: my life with a ludic system. *Personal and Ubiquitous Computing* 10 (2006), 60–65.
- [12] Mary L Gick. 2011. Singing, health and well-being: A health psychologist's review. *Psychomusicology: Music, Mind and Brain* 21, 1-2 (2011), 176.
- [13] Matthew J Hertenstein, Julie M Verkamp, Alyssa M Kerestes, and Rachel M Holmes. 2006. The communicative functions of touch in humans, nonhuman primates, and rats: a review and synthesis of the empirical research. *Genetic, social, and general psychology monographs* 132, 1 (2006), 5–94.
- [14] Maarten Houben, Rens Brankaert, Saskia Bakker, Gail Kenning, Inge Bongers, and Berry Eggen. 2020. The Role of Everyday Sounds in Advanced Dementia Care. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3313831.3376577
- [15] Yajing Hu, Yijun Zhao, Yaping Shao, Can Zhu, Jiayan Chen, Tianlei Shi, Zili Zhou, Fangtian Ying, Guanyun Wang, and Cheng Yao. 2021. Bubble Beats: A Breathing Exercise Game Based on Music Rhythm for Children. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference* (Athens, Greece) (IDC '21). Association for Computing Machinery, New York, NY, USA, 588–592. doi:10.1145/3459990.3465207
- [16] Chiara Jongerius, Roy S Hessels, Johannes A Romijn, Ellen MA Smets, and Marij A Hillen. 2020. The measurement of eye contact in human interactions: a scoping review. *Journal of Nonverbal Behavior* 44 (2020), 363–389.
- [17] Chris L Kleinke. 1986. Gaze and eye contact: a research review. *Psychological bulletin* 100, 1 (1986), 78.
- [18] Cun Li, Jun Hu, Bart J Hengeveld, and Caroline C.M. Hummels. 2019. Designing for older adults'life storytelling through a tangible interactive device. In *Conference proceedings of the Academy for design Innovation Management: Research perspectives in the era of transformations*, Vol. 2. Academy for Design Innovation Management, London, UK, 15–27. doi:10.33114/adim.2019.02_273 Academy for Design Innovation Management ; Conference date: 19-06-2019 Through 21-06-2019.
- [19] Cun Li, Linghao Zhang, Xu Lin, Kai Kang, Jun Hu, Bart Hengeveld, and Caroline Hummels. 2023. Towards the Senior Resident Social Interaction System: A Case Study of Interactive Gallery. *Systems* 11, 4 (2023), 204.
- [20] Xu Lin, Kai Kang, Cun Li, Jun Hu, Bart Hengeveld, Matthias Rauterberg, and Caroline Hummels. 2016. ViewBricks: a participatory system to increase social connectedness for the elderly in care homes. In *Intelligent Environments 2016*. IOS Press, Netherlands, 376–385. doi:10.3233/978-1-61499-690-3-376
- [21] Micah Linnemeier, Yi-Ying Lin, Gierad Laput, and Ramachandra Vijjapurapu. 2012. StoryCubes: connecting elders in independent living through storytelling. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI EA '12). Association for Computing Machinery, New York, NY, USA, 1321–1326. doi:10.1145/2212776.2212447
- [22] Viktor Müller and Ulman Lindenberger. 2011. Cardiac and respiratory patterns synchronize between persons during choir singing. *PLoS one* 6, 9 (2011), e24893.
- [23] Loveday Newman, Charlotte Stoner, and Aimee Spector. 2021. Social networking sites and the experience of older adult users: a systematic review. *Ageing & Society* 41, 2 (2021), 377–402.
- [24] María Ospina, Kenneth Bond, Mohammad Karkhaneh, Lisa Tjosvold, Ben Vandermeer, Yuanyuan Liang, Liza Bialy, Nicola Hooton, Nina Buscemi, Donna Dryden, and Terry Klassen. 2007. Meditation Practices for Health: State of the Research. *Evidence report/technology assessment* 155 (07 2007), 1–263.
- [25] Richard V Palumbo, Marisa E Marraccini, Lisa L Weyandt, Oliver Wilder-Smith, Heather A McGee, Siwei Liu, and Matthew S Goodwin. 2017. Interpersonal autonomic physiology: A systematic review of the literature. *Personality and social psychology review* 21, 2 (2017), 99–141.
- [26] Mirjana Prpa, Ekaterina R. Stepanova, Thecla Schiphorst, Bernhard E. Riecke, and Philippe Pasquier. 2020. Inhaling and Exhaling: How Technologies Can Perceptually Extend our Breath Awareness. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3313831.3376183
- [27] Michiko Sakaki, Ayano Yagi, and Kou Murayama. 2018. Curiosity in old age: A possible key to achieving adaptive aging. *Neuroscience & Biobehavioral Reviews* 88 (2018), 106–116.
- [28] Petr Slovák, Joris Janssen, and Geraldine Fitzpatrick. 2012. Understanding heart rate sharing: towards unpacking physiosocial space. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 859–868. doi:10.1145/2207676.2208526
- [29] Lynn Somerstein. 2010. Together in a room to alleviate anxiety: Yoga breathing and psychotherapy. *Procedia-Social and Behavioral Sciences* 5 (2010), 267–271.
- [30] Tobias Sonne and Mads Møller Jensen. 2016. ChillFish: A Respiration Game for Children with ADHD. In *Proceedings of the TEI '16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction* (Eindhoven, Netherlands) (TEI '16). Association for Computing Machinery, New York, NY, USA, 271–278. doi:10.1145/2839462.2839480
- [31] Richard M Suzman, David P Willis, and Kenneth G Manton. 1992. *The Oldest Old*. Oxford University Press, New York, NY, USA. doi:10.1093/oso/9780195050608.001.0001
- [32] Morton Wiener, Shannon Devoe, Stuart Rubinow, and Jesse Geller. 1972. Non-verbal behavior and nonverbal communication. *Psychological review* 79, 3 (1972), 185.